

# CPR E 4890: Computer Networking and Data Communication

## Lab Experiment #1 – Networking Utility Programs

### Summary:

This networking lab provided me with hands-on experience on essential diagnostic and analysis tools used in computer networks. I was able to learn how to identify network interfaces and their IP addresses, understand the loopback interface's role in troubleshooting, and measure network latency using ping commands to various destinations. I also discovered why local pings are significantly faster than remote ones due to differences between testing the internal network stack versus routing through multiple physical devices. The lab covered DNS resolution using nslookup to translate domain names to IP addresses, identify mail exchangers, and distinguish between aliases and canonical names.

Using traceroute and tcptraceroute, which tracked packet paths across networks and I was also able to learn more about why TCP packets often succeed where ICMP packets are dropped by routers. The iperf tool taught bandwidth measurement and unit conversion, while nmap introduced port scanning to identify open services. Finally, as for the packet capture tools like tcpdump and Wireshark, these enabled me to analyze network traffic in real-time, identify attack sources during simulated ping floods, examine TCP connection details, understand port usage for common protocols like HTTP and HTTPS, and differentiate between packet types (ICMP, UDP, TCP) used by various networking utilities. Overall, the lab equipped me with fundamental skills for network troubleshooting, performance optimization, and security monitoring.

### **ip a & ifconfig**

#### **1. What devices (interfaces) are available on your machine? What are the IP addresses assigned to them?**

There are 4 devices (interfaces) available on the machine. These are:

| Interface | IP Address     |
|-----------|----------------|
| lo        | 127.0.0.1      |
| enp4s0f0  | 192.168.254.22 |
| enp4s0f1  | 192.168.75.22  |
| enp0s31f6 | None           |

```
[489labuser@co2061-22 ~]$ ip -brief addr show
lo                UNKNOWN          127.0.0.1/8 ::1/128
enp4s0f0          UP                192.168.254.22/24 fe80::e63d:1aff:fea0:31b6/64
enp4s0f1          UP                192.168.75.22/24  fe80::e63d:1aff:fea0:31b7/64
enp0s31f6         DOWN
```

**2. What is the device (interface) named `lo` used for? Use manual pages or search online. What address is assigned to the device?**

The device (interface) named `lo` is called a loopback address and it is a reserved IPv4 address range 127.0.0.0/8. The loopback interface usually uses the first address 127.0.0.1 in this range. A loopback interface is a virtual interface and does not represent any hardware devices. It is just part of the internal TCP/IP stack of the OS. Our machine can use this interface to communicate with itself, and it's very useful in troubleshooting and diagnostics. The key purpose of the loopback address is to help in troubleshooting scenarios where we want to isolate and identify if a network issue is caused by an internal problem inside the machine network stack itself.

## ping

**3. Ping the following destinations and report the average round-trip-time (the latency) to each one:**

[www.google.com](http://www.google.com) : Takes an average of 16.070s

```
[489labuser@co2061-22 ~]$ ping -c 3 www.google.com
PING www.google.com (142.251.186.104) 56(84) bytes of data.
64 bytes from lcdfwe-in-f104.1e100.net (142.251.186.104): icmp_seq=1 ttl=100 time=16.0 ms
64 bytes from lcdfwe-in-f104.1e100.net (142.251.186.104): icmp_seq=2 ttl=100 time=15.9 ms
64 bytes from lcdfwe-in-f104.1e100.net (142.251.186.104): icmp_seq=3 ttl=100 time=16.3 ms

--- www.google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 15.874/16.070/16.295/0.173 ms
```

[www.iastate.edu](http://www.iastate.edu) : Takes an average of 28.578s

```
[489labuser@co2061-22 ~]$ ping -c 3 www.iastate.edu
PING www.iastate.edu (52.252.224.122) 56(84) bytes of data.
64 bytes from 52.252.224.122 (52.252.224.122): icmp_seq=1 ttl=109 time=28.5 ms
64 bytes from 52.252.224.122 (52.252.224.122): icmp_seq=2 ttl=109 time=28.8 ms
64 bytes from 52.252.224.122 (52.252.224.122): icmp_seq=3 ttl=109 time=28.4 ms

--- www.iastate.edu ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 28.421/28.578/28.778/0.148 ms
```

vm.iastate.edu : Takes an average of 0.940s

```
[489labuser@co2061-22 ~]$ ping -c 3 vm.iastate.edu
PING cvmweb.cvm.iastate.edu (129.186.162.8) 56(84) bytes of data.
64 bytes from cvmweb.cvm.iastate.edu (129.186.162.8): icmp_seq=1 ttl=60 time=0.761 ms
64 bytes from cvmweb.cvm.iastate.edu (129.186.162.8): icmp_seq=2 ttl=60 time=1.01 ms
64 bytes from cvmweb.cvm.iastate.edu (129.186.162.8): icmp_seq=3 ttl=60 time=1.05 ms

--- cvmweb.cvm.iastate.edu ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.761/0.940/1.049/0.127 ms
```

127.0.0.1 : Takes an average of 0.041s

```
[489labuser@co2061-22 ~]$ ping -c 3 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.036 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.044 ms

--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2071ms
rtt min/avg/max/mdev = 0.036/0.041/0.045/0.004 ms
```

[www.cam.ac.uk](http://www.cam.ac.uk) : Takes an average of 14.920s

```
[489labuser@co2061-22 ~]$ ping -c 3 www.cam.ac.uk
PING www.cam.ac.uk (23.185.0.3) 56(84) bytes of data.
64 bytes from 23.185.0.3 (23.185.0.3): icmp_seq=1 ttl=51 time=14.6 ms
64 bytes from 23.185.0.3 (23.185.0.3): icmp_seq=2 ttl=51 time=15.2 ms
64 bytes from 23.185.0.3 (23.185.0.3): icmp_seq=3 ttl=51 time=15.0 ms

--- www.cam.ac.uk ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 14.561/14.920/15.158/0.258 ms
```

4. Rank the latency from least to greatest. Why do you suppose this is the case?  
Why are pings to 127.0.0.1 so much faster than pings to other destinations? Use the internet, manual pages, or ask your TA if stuck.

| Name            | Time    |
|-----------------|---------|
| 127.0.0.1       | 0.041s  |
| vm.iastate.edu  | 0.940s  |
| www.cam.ac.uk   | 14.920s |
| www.google.com  | 16.070s |
| www.iastate.edu | 28.578s |

The reason why the latency for the ping command is the fastest to 127.0.0.1 compared to other destinations is because a ping to 127.0.0.1 tests the local machine's network stack

whereas for other destinations the request is being sent out through multiple routers, switches and long distances such as to servers.

5. Using the manual pages for ping (man ping), determine which flag/option can be used to specify the interface pings will be sent out of? Which option can be used to specify packet size?

Use this flag to specify the interface pings will be sent out of : -I

Use this flag to specify the packet size: -s

## traceroute & tcptraceroute

6. Perform a traceroute to google.com. How many hops does it take for your packets to reach google.com? What is the latency? About how many hops are within the ISU network (example internal address: vm.iastate.edu)?

```
[489labuser@co2061-22 ~]$ traceroute google.com
traceroute to google.com (142.250.114.100), 30 hops max, 60 byte packets
 1 _gateway (192.168.254.254) 0.233 ms 0.199 ms 0.177 ms
 2 router-129-186-5-0.tele.iastate.edu (129.186.5.254) 0.533 ms 0.512 ms 0.643 ms
 3 b31-mlps-p-hu1-0-18--to--b11-c12-pe-01-foul-0-33.tele.iastate.edu (129.186.0.65) 1.722 ms 1.703 ms e63-mlps-p-hu1-0-18--to--b11-c12-pe-01-foul-0-34.tele.iastate.edu (129.186.0.69) 0.643 ms
 4 b31-mlps-fpe-eth2-10--to--e63-mlps-p-hu0-2-0-1.tele.iastate.edu (129.186.0.141) 0.818 ms 0.943 ms b31-mlps-fpe-eth1-10--to--b31-mlps-p-hu0-2-0-1.tele.iastate.edu (129.186.0.135) 1.382 ms
 5 e63fr--b31fpe-vrf-data.tele.iastate.edu (129.186.254.245) 1.049 ms 0.887 ms 0.864 ms
 6 b31be-eth1-2.fusion.tele.iastate.edu (192.188.159.227) 0.864 ms e63be-eth1-2.fusion.tele.iastate.edu (192.188.159.229) 0.743 ms b31be-eth1-2.fusion.tele.iastate.edu (192.188.159.227) 0.716 ms
 7 routera-192-188-159-96.tele.iastate.edu (192.188.159.100) 0.661 ms 0.787 ms 0.773 ms
 8 rtr-b31be-vlan933.tele.iastate.edu (192.188.159.105) 1.506 ms 1.428 ms 1.118 ms
 9 rtr-e63isp1-be156.tele.iastate.edu (192.188.159.157) 2.937 ms 3.211 ms 3.633 ms
10 rtr-b31isp1-be162.tele.iastate.edu (192.188.159.162) 3.642 ms 2.326 ms 3.141 ms
11 fourhundredge-0-0-0-10-1421.core2.kans.net.internet2.edu (198.71.47.103) 8.431 ms 8.417 ms 8.608 ms
12 fourhundredge-0-0-0-1-4079.core2.dall.net.internet2.edu (163.253.2.10) 15.805 ms 15.881 ms 15.158 ms
13 fourhundredge-0-0-0-52.4079.aggr2.dall3.net.internet2.edu (163.253.2.89) 17.054 ms 17.040 ms 17.026 ms
14 * * *
15 * * *
16 142.251.60.46 (142.251.60.46) 15.380 ms 72.14.234.182 (72.14.234.182) 15.522 ms 142.251.76.36 (142.251.76.36) 15.168 ms
17 216.239.59.72 (216.239.59.72) 15.507 ms 142.251.194.194 (142.251.194.194) 14.824 ms 15.192 ms
18 192.178.99.221 (192.178.99.221) 16.682 ms 142.251.249.97 (142.251.249.97) 15.148 ms 192.178.97.207 (192.178.97.207) 17.129 ms
19 142.251.225.233 (142.251.225.233) 16.473 ms 172.253.176.159 (172.253.176.159) 16.266 ms 172.253.176.155 (172.253.176.155) 17.092 ms
20 142.251.196.47 (142.251.196.47) 16.759 ms 16.740 ms 192.178.255.136 (192.178.255.136) 16.240 ms
21 209.85.143.223 (209.85.143.223) 15.823 ms 142.250.224.23 (142.250.224.23) 15.915 ms 142.250.224.13 (142.250.224.13) 15.955 ms
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *
```

It takes 16 hops to get to google.com. This was checked using an online IP lookup tool and it was found that on hop #16 (the IP stated is google's). The latency was roughly 15ms. There are about 10 hops within the ISU network.

7. Perform another traceroute to google.com using TCP packets instead of ICMP. Are there any differences in the output? (Hint: sudo is required for tcptraceroute)

```
[489labuser@co2061-22 ~]$ sudo tcptraceroute google.com

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for 489labuser:
Running:
traceroute -T -O info.google.com
traceroute to google.com (142.250.114.113), 30 hops max, 60 byte packets
 1 _gateway (192.168.254.254)  0.528 ms  0.509 ms  0.499 ms
 2 router-129-186-5-0.tele.iastate.edu (129.186.5.254)  0.972 ms  0.964 ms  1.042 ms
 3 e63-mlps-p-hul-0-18--to--b11-cl2-pe-01-foul-0-34.tele.iastate.edu (129.186.0.69)  0.932 ms b31-mlps-p-hul-0-19--to--b11-cl2-pe-01-foul-0-34.tel
e.iastate.edu (129.186.0.69)  0.915 ms
 4 b31-mlps-fpe-eth2-10--to--e63-mlps-p-hul-0-2-0-1.tele.iastate.edu (129.186.0.141)  1.189 ms *  1.276 ms
 5 e63fr--b31fpe-vrf-data.tele.iastate.edu (129.186.254.245)  0.983 ms b31fr--b31fpe-vrf-data.tele.iastate.edu (129.186.254.255)  1.068 ms e63fr--b31fpe-vrf-data.tele.iastate.edu (129.186.254.245)  0.963 ms
 6 e63be-eth2-2.fusion.tele.iastate.edu (192.188.159.231)  1.048 ms * *
 7 * * router-192-188-159-96.tele.iastate.edu (192.188.159.100)  0.603 ms
 8 rtr-e63be-vlan933.tele.iastate.edu (192.188.159.100)  1.074 ms * *
 9 rtr-e63isp1-bel54.tele.iastate.edu (192.188.159.155)  2.389 ms  2.123 ms  1.798 ms
10 rtr-b31isp1-bel62.tele.iastate.edu (192.188.159.162)  2.245 ms  2.677 ms  2.906 ms
11 fourhundredge-0-0-0-10.1421.core2.kans.net.internet2.edu (198.71.47.103)  9.152 ms  9.142 ms  9.119 ms
12 fourhundredge-0-0-0-1.4079.core2.dall.net.internet2.edu (163.253.2.10)  16.872 ms  18.070 ms  18.058 ms
13 fourhundredge-0-0-0-52.4079.aggr1.dall3.net.internet2.edu (163.253.2.87)  17.694 ms fourhundredge-0-0-0-52.4079.aggr2.dall3.net.internet2.edu (163.253.2.89)  18.203 ms  16.170 ms
14 * * *
15 108.170.231.46 (108.170.231.46)  14.680 ms 108.170.225.147 (108.170.225.147)  17.154 ms  15.151 ms
16 142.251.237.112 (142.251.237.112)  16.033 ms 142.250.60.238 (142.250.60.238)  15.587 ms 172.253.176.28 (172.253.176.28)  14.818 ms
17 192.178.97.207 (192.178.97.207)  16.986 ms 192.178.98.25 (192.178.98.25)  16.952 ms 188.170.228.73 (188.170.228.73)  15.205 ms
18 172.253.176.157 (172.253.176.157)  16.696 ms 142.251.194.60 (142.251.194.60)  16.892 ms 216.239.58.83 (216.239.58.83)  16.773 ms
19 209.85.252.234 (209.85.252.234)  16.423 ms 108.170.233.60 (108.170.233.60)  15.892 ms 142.251.196.89 (142.251.196.89)  16.653 ms
20 216.239.63.153 (216.239.63.153)  16.316 ms 142.250.224.25 (142.250.224.25)  16.305 ms 209.85.143.223 (209.85.143.223)  16.692 ms
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * rrr-in-f113.1e100.net (142.250.114.113) <syn,ack>  16.067 ms *
```

The key difference between the traceroute command we ran, and the sudo tcptraceroute is the routers may be configured to drop/not respond to specific packets (namely our ICMP packets). However, when we use sudo tcptraceroute, our packets being used are now TCP SYN packets instead of ICMP packets. We can notice that there are less asterisks when using TCP packets instead which goes to show that our ICMP packets were being dropped by some routers.

## nslookup

### 8. Perform nslookup of www.iastate.edu. What is the IP address of this server?

```
[489labuser@co2061-22 ~]$ nslookup www.iastate.edu
Server:      129.186.1.200
Address:     129.186.1.200#53

Name:   www.iastate.edu
Address: 52.252.224.122
```

After doing nslookup of [www.iastate.edu](http://www.iastate.edu), the IP address for the server comes out to 129.186.1.200

### 9. What is the mail exchanger for ece.iastate.edu?

```
[489labuser@co2061-22 ~]$ nslookup -query=MX ece.iastate.edu
Server:      129.186.1.200
Address:     129.186.1.200#53

ece.iastate.edu mail exchanger = 10 vulcan.ece.iastate.edu.
```

The mail exchanger for ece.iastate.edu is vulcan.ece.iastate.edu.

**10. Which of the following are aliases? If it's an alias, record its canonical name.**

[www.apple.com](http://www.apple.com) : This is an alias, it's canonical name is www-apple-com.v.aapling.com.

Apple.com : This is not an alias. It is the canonical name.

[www.google.com](http://www.google.com) : This is not an alias. It is a canonical name.

## iperf

**11. Your TA is running the iperf server program on the machine 192.168.254.1. Run the iperf client program against the TA's iperf server and report the bandwidth of the connection in Gigabits per second (Gbps), Gigabytes per second (GBps), Megabits per second (Mbps), and Megabytes per second (MBps).**

```
[489labuser@co2061-22 ~]$ iperf -c 192.168.254.1
-----
Client connecting to 192.168.254.1, TCP port 5001
TCP window size: 85.0 KByte (default)
-----
[  1] local 192.168.254.22 port 58030 connected with 192.168.254.1 port 5001
[ ID] Interval      Transfer    Bandwidth
[  1] 0.00-10.03 sec  1.10 GBytes  940 Mbits/sec
```

After running the iperf command on the server program being hosted by the TA, the reported bandwidth is 940 Mbps, which is also 0.94 Gbps, 117.5 MBps, 0.1175 GBps.

## nmap

**12. Run a port map against your own machine. Is port 22 open? Which service runs on this port?**

```
[489labuser@co2061-22 ~]$ nmap -Pn 127.0.0.1
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-27 14:46 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000026s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp    open  rpcbind
631/tcp    open  ipp

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
```

Port 22 is open. Secure Shell is the service that runs on port 22.

## tcpdump

13. For this exercise, work with a partner and alternate roles. Your machine is currently undergoing a ping flood attack! On partner #1's computer, execute `ping [address]` where "address" is partner #2's IP address. On partner #2's computer, use `tcpdump` and filter for ICMP packets to determine the IP address of the machine that is sending the packets. Include the IP addresses of your machine and your partner's machine in your report. (Note: use IP address 192.168.254.x where x can be found using earlier command; also, for a more accurate flood attack, `-f` can be used but requires `sudo`).

```
14:55:07.133175 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 254, length 64
14:55:07.133178 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 254, length 64
14:55:07.133293 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 255, length 64
14:55:07.133296 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 255, length 64
14:55:07.133411 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 256, length 64
14:55:07.133414 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 256, length 64
14:55:07.133530 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 257, length 64
14:55:07.133533 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 257, length 64
14:55:07.133648 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 258, length 64
14:55:07.133651 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 258, length 64
14:55:07.133767 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 259, length 64
14:55:07.133770 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 259, length 64
14:55:07.133887 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 260, length 64
14:55:07.133890 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 260, length 64
14:55:07.133991 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 261, length 64
14:55:07.133994 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 261, length 64
14:55:07.134109 IP 192.168.254.21 > co2061-22.ece.iastate.edu: ICMP echo request, id 6, seq 262, length 64
14:55:07.134112 IP co2061-22.ece.iastate.edu > 192.168.254.21: ICMP echo reply, id 6, seq 262, length 64
^C
262808 packets captured
263474 packets received by filter
```

The above is the output I got when my machine was undergoing the ping flood attack. My partner's IP address was 192.168.254.21 and my machine's IP address 192.168.254.22.

Below is the screenshot for when I carried out the attack against my partner.

```
[489labuser@co2061-22 ~]$ sudo ping -f 192.168.254.21
PING 192.168.254.21 (192.168.254.21) 56(84) bytes of data.
.^C
--- 192.168.254.21 ping statistics ---
33571 packets transmitted, 33570 received, 0.00297876% packet loss, time 3895ms
rtt min/avg/max/mdev = 0.076/0.102/0.525/0.005 ms, ipg/ewma 0.116/0.100 ms
```

## Wireshark

### 14. For each connection, indicate the duration and the number of packets sent from your computer during that connection.

| Address A      | Port A | Address B      | Port B | Packets | Bytes | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B |
|----------------|--------|----------------|--------|---------|-------|---------------|-------------|---------------|-------------|-----------|----------|--------------|
| 192.168.254.22 | 39456  | 52.252.224.122 | 80     | 32      | 2947  | 17            | 1499        | 15            | 1448        | 30.975362 | 115.8975 | 103          |
| 192.168.254.22 | 39462  | 52.252.224.122 | 80     | 6       | 424   | 4             | 280         | 2             | 144         | 30.983395 | 5.6670   | 395          |
| 192.168.254.22 | 56362  | 52.252.224.122 | 443    | 514     | 1146k | 245           | 20k         | 269           | 1125k       | 31.040827 | 170.5874 | 965          |
| 192.168.254.22 | 56366  | 52.252.224.122 | 443    | 63      | 50k   | 30            | 3225        | 33            | 47k         | 31.218642 | 170.4095 | 151          |
| 192.168.254.22 | 56382  | 52.252.224.122 | 443    | 42      | 12k   | 24            | 3172        | 18            | 9078        | 32.114387 | 170.5179 | 148          |
| 192.168.254.22 | 56394  | 52.252.224.122 | 443    | 25      | 8945  | 15            | 1734        | 10            | 7211        | 32.127666 | 0.2050   | 67k          |

52.252.224.122 is the [www.iastate.edu](http://www.iastate.edu) IP address when looking through nslookup. In the above screenshot, we can see that the duration and # of packets sent from my computer during the first connection is 115.8975 and 17 packets sent (A -> B) respectively. In the second connection the duration is 5.7770 and 4 packets sent (A -> B). In the third connection the duration is 170.5874 and 245 packets sent (A -> B). In the fourth connection the duration is 170.4095 and 30 packets sent (A -> B). In the fifth connection the duration is 170.5179 and 24 packets sent (A -> B). In the last connection, the duration is 0.2050 and 15 packets sent (A -> B).

### 15. What were the destination ports used and what protocols do they correspond to?

The destination ports used were 80 and 443 respectively. 80 is HTTP and 443 is HTTPS. The ports used from our machine were ephemeral ports.

### 16. Determine how much data (in bytes) each ICMP packet contains. (Hint: the conversations panel does not support ICMP explicitly; you can capture the traffic between your two nodes to make sure the result is valid)

An ICMP packet contains 98 bytes in each packet. I was able to find this out by pinging my own localhost and changing the interface on Wireshark to loopback.

|    |              |           |           |      |                        |                                                |
|----|--------------|-----------|-----------|------|------------------------|------------------------------------------------|
| 1  | 0.000000000  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=24/6144, ttl=64 (reply in 2)    |
| 2  | 0.000010025  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=24/6144, ttl=64 (request in 1)  |
| 3  | 1.024076214  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=25/6400, ttl=64 (reply in 4)    |
| 4  | 1.024092342  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=25/6400, ttl=64 (request in 3)  |
| 5  | 2.048084086  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=26/6656, ttl=64 (reply in 6)    |
| 6  | 2.048098405  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=26/6656, ttl=64 (request in 5)  |
| 7  | 3.072226952  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=27/6912, ttl=64 (reply in 8)    |
| 8  | 3.072244890  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=27/6912, ttl=64 (request in 7)  |
| 9  | 4.096245369  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=28/7168, ttl=64 (reply in 10)   |
| 10 | 4.096262762  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=28/7168, ttl=64 (request in 9)  |
| 11 | 5.120216575  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=29/7424, ttl=64 (reply in 12)   |
| 12 | 5.120234572  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=29/7424, ttl=64 (request in 11) |
| 13 | 6.144207763  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=30/7680, ttl=64 (reply in 14)   |
| 14 | 6.144229404  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=30/7680, ttl=64 (request in 13) |
| 15 | 7.168007739  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=31/7936, ttl=64 (reply in 16)   |
| 16 | 7.168026692  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=31/7936, ttl=64 (request in 15) |
| 17 | 8.192240810  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=32/8192, ttl=64 (reply in 18)   |
| 18 | 8.192257594  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=32/8192, ttl=64 (request in 17) |
| 19 | 9.216283273  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=33/8448, ttl=64 (reply in 20)   |
| 20 | 9.216299934  | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=33/8448, ttl=64 (request in 19) |
| 21 | 10.240148076 | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=34/8704, ttl=64 (reply in 22)   |
| 22 | 10.240164926 | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=34/8704, ttl=64 (request in 21) |
| 23 | 11.264014566 | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=35/8960, ttl=64 (reply in 24)   |
| 24 | 11.264029575 | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=35/8960, ttl=64 (request in 23) |
| 25 | 12.288105426 | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) request | id=0x0000, seq=36/9216, ttl=64 (reply in 26)   |
| 26 | 12.288123504 | 127.0.0.1 | 127.0.0.1 | ICMP | 98 Echo (ping) reply   | id=0x0000, seq=36/9216, ttl=64 (request in 25) |



17. What types of packets are sent with traceroute? What types of packets are sent with tcptraceroute?

|    |              |                |               |     |                         |
|----|--------------|----------------|---------------|-----|-------------------------|
| 66 | 22.996426879 | 192.168.254.22 | 23.38.109.150 | UDP | 76 38756 → 33434 Len=32 |
| 67 | 22.996462905 | 192.168.254.22 | 23.38.109.150 | UDP | 76 43057 → 33435 Len=32 |
| 68 | 22.996485874 | 192.168.254.22 | 23.38.109.150 | UDP | 76 46153 → 33436 Len=32 |
| 69 | 22.996506777 | 192.168.254.22 | 23.38.109.150 | UDP | 76 47939 → 33437 Len=32 |
| 70 | 22.996529213 | 192.168.254.22 | 23.38.109.150 | UDP | 76 37731 → 33438 Len=32 |
| 71 | 22.996551210 | 192.168.254.22 | 23.38.109.150 | UDP | 76 46604 → 33439 Len=32 |
| 72 | 22.996572434 | 192.168.254.22 | 23.38.109.150 | UDP | 76 49101 → 33440 Len=32 |
| 73 | 22.996593019 | 192.168.254.22 | 23.38.109.150 | UDP | 76 41778 → 33441 Len=32 |
| 74 | 22.996613705 | 192.168.254.22 | 23.38.109.150 | UDP | 76 57835 → 33442 Len=32 |
| 75 | 22.996633695 | 192.168.254.22 | 23.38.109.150 | UDP | 76 48515 → 33443 Len=32 |
| 76 | 22.996655322 | 192.168.254.22 | 23.38.109.150 | UDP | 76 55770 → 33444 Len=32 |
| 77 | 22.996675714 | 192.168.254.22 | 23.38.109.150 | UDP | 76 46343 → 33445 Len=32 |

The packets sent via traceroute are UDP packets when we visit [www.ebay.com](http://www.ebay.com).

|    |              |                |               |     |                                                                                             |
|----|--------------|----------------|---------------|-----|---------------------------------------------------------------------------------------------|
| 67 | 11.004614915 | 192.168.254.22 | 23.38.109.136 | TCP | 76 57137 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 68 | 11.004628021 | 192.168.254.22 | 23.38.109.136 | TCP | 76 33137 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 69 | 11.004637874 | 192.168.254.22 | 23.38.109.136 | TCP | 76 46783 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 70 | 11.004647801 | 192.168.254.22 | 23.38.109.136 | TCP | 76 36751 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 71 | 11.004657391 | 192.168.254.22 | 23.38.109.136 | TCP | 76 56899 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 72 | 11.004666344 | 192.168.254.22 | 23.38.109.136 | TCP | 76 40997 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 73 | 11.004670515 | 192.168.254.22 | 23.38.109.136 | TCP | 76 41473 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 74 | 11.004685262 | 192.168.254.22 | 23.38.109.136 | TCP | 76 46671 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 75 | 11.004694405 | 192.168.254.22 | 23.38.109.136 | TCP | 76 48231 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |
| 76 | 11.004704652 | 192.168.254.22 | 23.38.109.136 | TCP | 76 58699 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3886693626 TSecr=0 WS=4 |

The packets sent via tcptraceroute are TCP packets when we visit [www.ebay.com](http://www.ebay.com).